SRI CHAITANYA EDUCATIONAL INSTITUTIONS

A.P,TELANGANA,KARNATAKA,TAMILNADU,MAHARASHTRA,DELHI,RANCHI,CHANDIGARH

NEET GRAND TEST-10

Name :..... Hall Ticket No: | | | | | | | |

INSTRUCTIONS TO CANDIDATES

1. The Model **NEET- 2019** is of **3 Hrs** duration. **Time: 2 PM TO 5 PM**
2. The question paper for NEET-2019 consists of 180 questions comprising 45 questions in Botany, 45 in Zoology, 45 in Physics and 45 in Chemistry for NEET.
3. All questions are of objective type (Multiple choices only)
4. Each question carries four marks.
5. Negative marking: one mark will be deducted for every wrongly answered question.
6. Total Marks 720.
7. The candidates are prohibited from carrying any paper to the examination hall except **HALL TICKET**.
8. No Calculators, Mini-Cards, Watches with Calculators, Pager, Cell Phone, Slide rules or outer aids to calculation will be allowed in the examination hall.
9. Candidates once admitted will not be allowed to leave the hall till half an hour before the closing of the test.
10. A separate sheet is attached in the middle of this booklet for rough work, you can detach and use it.
11. A detachable answer sheet with 180 question blocks, with 4 circles corresponding to 4 multiple choice for each question will be provided. Use HB Pencil to darken the appropriate circle against the question number provided in the sheet. Answer should be marked only on the answer sheet, but not on the question paper booklet.

1. Strobili are formed in

(1) *Selaginella* and *Equisetum*

(2) *Sphagnum* and *Polytrichum*

(3) *Pinus* and female plant of *Cycas*

(4) *Agaricus* and *Ficus*

2. A plant having genotype AABB was crossed with another plant having genotype aabb. The F_1 plants produced gametes AB, Ab, aB and ab in the ratio of 1 : 1 : 1 : 1. Then

(1) Genes A and B can be present in nonhomologus chromosomes

(2) Genes A and B can be distantly present in same chromosome

(3) Genes A and B are showing independent assortment

(4) All of these statements are correct

3. Which of the following is advantage of tissue culture in plant breeding ?

(1) Production of somaclonal variants

(2) Rapid clonal propagation of desired plant

(3) To obtain virus free plants through meristem culture

(4) All of these

4. In ecological succession, in successive seral stages there is all except

(1) Increase in species diversity

(2) Invariabily increase in water content of substratum

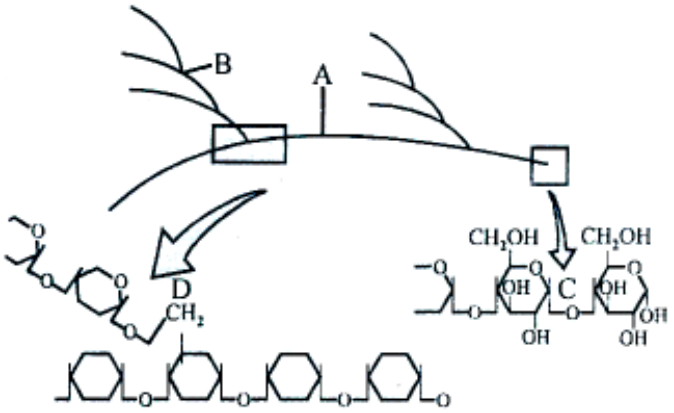
(3) Increase in total biomass

(4) Populations of some species increase whereas that of other species decline

5. Which one of the following organisms is scientifically correctly named, correctly printed according to the international rules of Nomenclature and correctly described?

(1)	*E.coli*	Full name *Entamoeba coli*, a commonly occurring bacterium in human intestine
(2)	*Mangifera indica*	Common name of Indian Mango, A monocotyledonae
(3)	*Felis tigris*	The Indian tiger, well protected in Gir forest
(4)	*Azolla pinnata*	Aquatic fern, show symbiotic association with prokaryotes

6. Give below is the diagrammatic sketch of certain type of polysaccharide. Identify the parts labelled A, B, C, D and select the right option about them



	Part A	Part B	Part C	Part D
(1)	Branching chain	Linear chain	a (1-6) glycosidic linkage	a (1-4) glycosidic linkage
(2)	Linear chain	Branching chain	a (1-4) glycosidic linkage	a (1-6) glycosidic linkage
(3)	Branching chain	a (1-6) glycosidic linkage	a (1-4) glycosidic linkage	Linear chain
(4)	Linear chain	Branching chain	a (1-6) glycosidic linkage	a (1-4) glycosidic linkage

7. A flower can be said to be a member of fabaceae if it shows

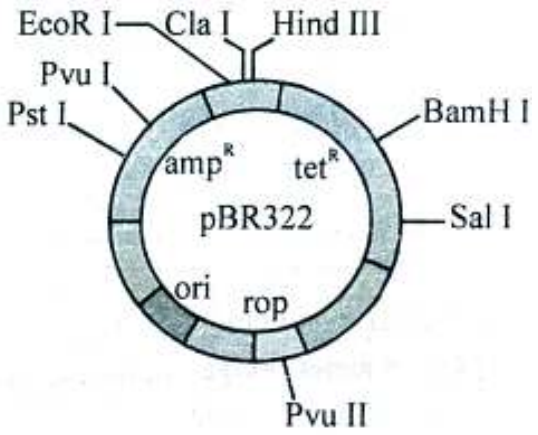
(1) Epipetalous stamens

(2) Twisted petals

(3) Standard, keel and wings

(4) Valvate petals

8. The figure given below is the diagrammatic representation of the vector pBR322. Which one of the given options correctly identifies its certain components ?



(1) Hind III, EcoRI-Antibiotic Resistance gene

(2) Amp^R, tet^R – Selectable markers

(3) Ori-Original restriction enzyme

(4) Rop – Reduced osmotic pressure

9. Choose the incorrect statement

(1) The vast majority of bacteria are heterotrophs

(2) Chemosynthetic bacteria are example of Autotrophs

(3) nif gene is present in all nitrifying bacteria

(4) Bacterial structure is very simple and they are very complex in behaviour

10. The conventional method for development of disease resistance are

(1) Mutation and selection

(2) Mutation and hybridisation

(3) Introduction and tissue culture

(4) Hybridisation and selection

11. Read the following term carefully

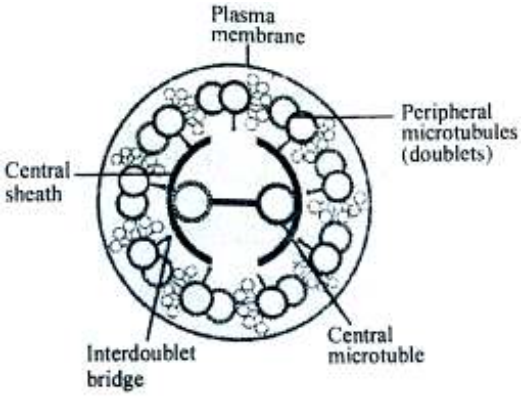
Cilia, Flagella, Heterocyst, Unicellular eukaryotes, Photosynthesis, Embryo, Zygote, Tissue, Seed, Pili, Pellicle, Chemosynthesis, Nitrogen fixation.

How many terms are only and only related with Monera and not found in any other kingdom.

(1) 8 (2) 7

(3) 4 (4) 10

12. Which of the following is not labeled correctly



(1) Plasma membrane

(2) Inter doublet bridge

(3) Central microtubule

(4) Central sheath

13. In respiratory pathways, which of the following dehydrogenase enzymes employ flavin nucleotide ?

(1) Glyceraldehyde-3-phosphate dehydrogenase and lactate dehydrogenase

(2) Pyruvate dehydrogenase and succinate dehydrogenase

(3) Succinate dehydrogenase and NADH dehydrogenase

(4) Succinate dehydrogenase and a – ketoglutarate dehydrogenase

14. Choose the incorrect statement from following

(1) Yeast are used to make bread and beer

(2) Fungi prefer to grow in warm and humid places

(3) Cell wall of fungi are composed of chitin and polysaccharides

(4) Fungi show symbiotic associated with algae as mycorrhiza and with roots of higher plants as lichens

15. Radial vascular bundles are found in

(1) Stem (2) Leaves

(3) Both (1) & (2) (4) Roots

16. Meiosis I does not have one of the events out of the four given below

(1) Bivalent chromosome align on the equatorial plate

(2) Crossing over occurs between the sister chromatid of homologus chromosomes

(3) Homologous chromosomes separate, while sister chromatids remain associated at their centromers

(4) Recombination nodule is formed

17. Morgan worked with *Drosophila* because

(i) They could be grown on simple medium

(ii) Complete their life cycle in about six weeks

(iii) Single mating could produce a large number of progeny flies

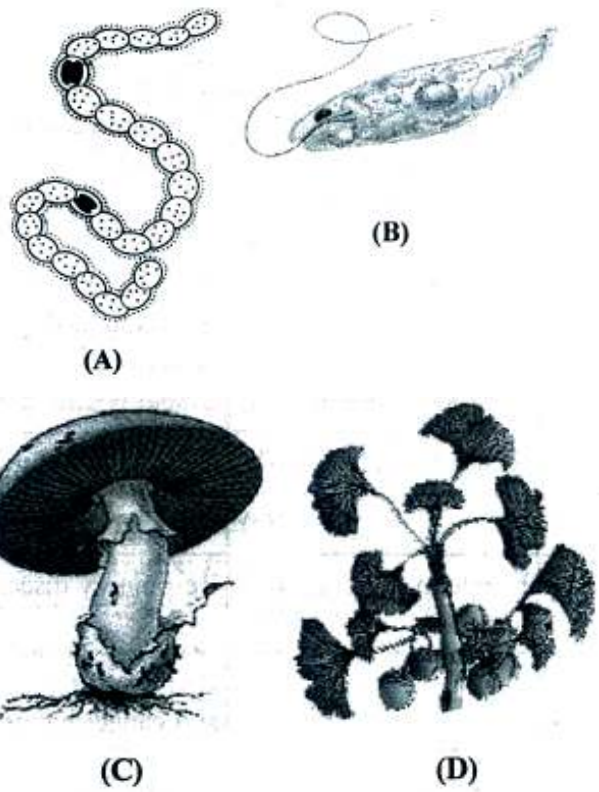
(iv) Clear differentiation of the sexes male (large size) and female (small size)

Choose the correct option

(1) i, ii, iv (2) i, iii

(3) i, iv (4) i, ii, iii, iv

18.



Identify the A, B, C and D

	A	B	C	D
(1)	*Nostoc*	*Paramecium*	*Agaricus*	*Sphagnum*
(2)	*Oscillatoria*	*Euglena*	*Puccinia*	*Ginkgo*
(3)	*Ulothrix*	*Euglena*	*Agaricus*	*Ginkgo*
(4)	*Nostoc*	*Euglena*	*Agaricus*	*Ginkgo*

19. Casparian strips are found on the

(1) Tangential and radial walls of pericycle cells of roots

(2) Tangential and radial walls of endodermal cells of roots

(3) Tangential and radial walls of hypodermal cells of stem

(4) Only radial walls of endodermal cells of roots

20. Micropropagation is

(1) Propagation of microbes in vitro

(2) Propagation of plant in vitro

(3) Growing plants on smaller scale

(4) Propagation of virus in vitro

21. Match the following.

	Column-I		Column-II
a)	**Jasmine**	**i)**	**Sucker**
b)	**Pineapple**	**ii)**	**Rhizome**
c)	**Zaminkand**	**iii)**	**Stolon**
d)	**Ginger**	**iv)**	**Corm**

	a	**b**	**c**	**d**
(1)	i	iii	iv	ii
(2)	ii	i	iv	iii
(3)	iii	i	ii	iv
(4)	iii	i	iv	ii

22. The scientific process by which crop plants are enriched with certain desirable nutrients is called

(1) Biolistic

(2) Bioprospecting

(3) Biofortification

(4) Bioremediation

23. Read the following statements regarding DNA polymerase used in PCR

a) It is isolated from thermosensitive bacteria

b) It remains active at high temperature

c) It is used to ligate introduced DNA in recipient cells

d) It serves as a selectable marker

How many statements are correct ?

(1) One (2) Two

(3) Three (4) Four

24. Which of the following statements does not hold true for restriction enzyme ?

(1) It recognise a palindromic nucleotide sequence

(2) It is an endonuclease

(3) It is isolated from bacteria

(4) It produces different kind of sticky ends in different DNA molecule

25. *Monascus purpureus* is a yeast used commercially in the production of

(1) Immunosuppressive drug

(2) Blood cholesterol lowering agent

(3) Ethanol

(4) Enzyme for removing clots from the blood vessels

26. Coconut fruit develops from

(1) Monocarpellary Superior ovary

(2) Monocarpellary Inferior ovary

(3) Multicarpellary Superior ovary

(4) Multicarpellary Inferior ovary

27. The vascular cambium of roots resembles the vascular cambium of stem in

(1) Its origin from ground tissue completely

(2) Its wavy outline

(3) Its function

(4) All of these

28. Choose the incorrect match from following

(1) Phycomycetes – *Albugo*

(2) Ascomycetes – *Alternaria*

(3) Basidiomycetes – *Ustilago*

(4) Deuteromycetes – *Trichoderma*

29. Which one of the following stage is correctly described with their event ?

(1)	Anaphase I	The bivalent chromosomes align on the equatorial plate
(2)	Diakinesis	Beginning of dissolution of synaptonemal complex
(3)	Metaphase	Spindle fibres attach to kinetochores of chromosomes
(4)	Zygotene	Apperance of recombinant nodule

30. Watson and Crick proposed certain features for double helical model of structure of DNA. Which of the following is not a feature of double helical structure of DNA ?

(1) Two chains are anti-parallel to each other

(2) Nitrogen bases of two strands are complementary and paired through covalent bonds

(3) Each DNA strand is made up of nucleotides

(4) Nitrogen bases are present towards inner side of double helix

31. Given below figure represent the stage of cell division. Read the following statements.



(i) Nucleolus, golgi complex and ER reform

(ii) Chromatids move to opposite pole

(iii) Activity of recombinase enzyme

(iv) Homologous chromosomes separate while sister chromatids associated at their centromere

(v) Initiation of the assembly of mitotic spindle

How many of the above statements are not true with respect to above figure

(1) Four (2) Three

(3) Five (4) Two

32. In which chromosome centromer is situated slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm.

(1) Metacentric (2) Submetacentric

(3) Telocentric (4) Acrocentric

33. Match the column-I with column-II

	Column-I		**Column-II**
a)	**Exine**	**i)**	**Pecto-cellulose**
b)	**Intine**	**ii)**	**Sporopollenin**
c)	***Calotropis***	**iii)**	**Pollinium**
d)	***Salvia***	**iv)**	**Turnpipe**

	a	**b**	**c**	**d**
(1)	iii	i	iv	ii
(2)	ii	i	iii	iv
(3)	i	iv	ii	iii
(4)	iii	ii	i	iv

34. In the inheritance of flower colours in dog flower plant, the F_1 had a phenotype that

(1) Resembles both of the parents

(2) Did not resembles either of the two parents

(3) Resembles with one parents

(4) Did not reappear in F_2

35. Based on observations on monohybrid crosses Mendel draw some conclusions. Which of the following is not correct ?

(1) Characters are controlled by discrete units called factors

(2) Factors occur in pairs

(3) In case two pairs of contrasting traits are together, one segrigates independent of the other

(4) The postulate of dominance also explains the proportion of 3 : 1 obtained at the F_2

36. The accessibility of promoter regions of prokaryotic DNA by RNA polymerase is in many cases regulated by the interaction of some protein with sequence termed as

(1) Promoter (2) Operator

(3) Regulator (4) Cistron

37. Find out correct statement :

(A) Synergids and antipodal degenerate after fertilization

(B) Polar nuclei produce diploid secondary nucleus

(C) Male gamete fuses with secondary nucleus to form zygote

(D) Male gamete fuses with egg cell to form triploid primary endosperm nucleus

(1) A and B (2) B and C

(3) A, B and D (4) A, C and D

38. Morgan coined the term _____ to describe the physical association of genes on a chromosome & the term _____ to describe the generation of non-parental gene combinations :

(1) Recombination ; Linkage

(2) Recombination; Non-recombination

(3) Linkage; Non-recombination

(4) Linkage; Recombination

39. Dominance is not an autonomous feature of a gene or the product, when

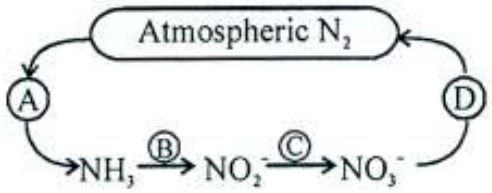
(1) More than one gene influence the same phenotype

(2) More than one phenotype is influenced by the same gene

(3) More than one genotype is influenced by the same gene

(4) More than one allele are there for a character

40.



Identify the different process in a part of nitrogen cycle represented by A, B, C and D find the correct match regarding the microbes that perform these processes :

(1)	A → *Rhizobium*; C → *Nitrosomonas*;	B → *Nitrobacter*; D → *Pseudomonas*
(2)	A → *Rhizobium*; C → *Nitrobacter*;	B → *Nitrosomonas*; D → *Pseudomonas*
(3)	A → *Frankia*; C → *Nitrobacter*;	B → *Nitrosomonas*; D → *Rhodospirillum*
(4)	A → *Chlorobium*; C → *Nitrobacter*;	B → *Nostoc*; D → *Rhodospirillum*

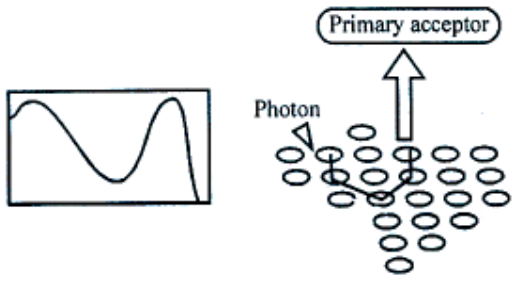
41. Zinc, Chlorine, Potassium, Sulphur, Manganese, Copper, Calcium, Magnesium.

How many of these are micronutrients?

(1) 5 (2) 3

(3) 4 (4) 6

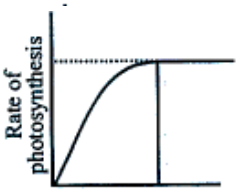
42. The four sketches (A, B, C and D) given below, represent four different phenomenons. Which one of these is correctly identified in the options given ?



(A) (B)

In	Out
Six CO_2	One glucose
18 ATP	18 ADP
12NADPH	12 NADP

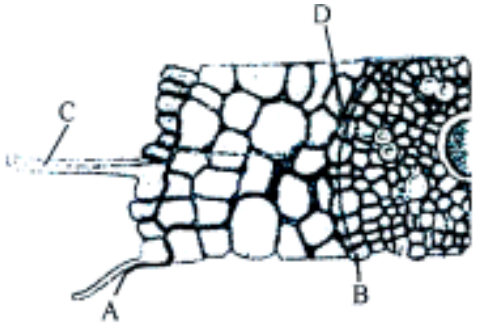
(C)



(D)
Light intensity

	A	B	C	D
(1)	Absorption spectrum of chlorophyll	Light harvesting complex	C_4 cycle balance sheet	Light saturation point
(2)	Action spectrum of photosynthesis	Cycle photo-phosph-orylation	C_3 cycle balance sheet	Light compensation point
(3)	Action spectrum of photo-synthesis	Light harvesting	C_3 cycle balance sheet	Light compensation point
(4)	Action spectrum of photo-synthesis	Light harvesting complex	Calvin cycle balance sheet	Light saturation point

43. Given below is the diagrammatic sketch of certain phenomenon. Identify A, B, C, D and select the right about them :



	A	B	C	D
(1)	Symplastic path	Endodermis	Apoplastic path	Casparian strip
(2)	Apoplastic path	Casparian strip	Symplastic path	Pericycle
(3)	Apoplastic path	Casparian strip	Symplastic path	Endodermis
(4)	Symplastic path	Casparian strip	Apoplastic path	Endodermis

44. A phenomenon that helps some plants in minimising water loss while maximising the availability of CO_2 is

(1) To keep their stomata open during night

(2) Operation of C_4 photosynthetic pathway

(3) Closure of stomata during water deficiency

(4) Non operation of photorespiration at low temperature

45. In the options given below A, B, C and D represent various function regulated by phytohormones. Choose the option that shows all the functions regulated by same hormone

(1) A-Production of new leaves, B-Overcomes apical dominance, C-Promotes nutrient mobilization, D-Apical hook formation

(2) A-Delaying of senescence, B-Speeds up malting, C-Early seed production in conifers, D-Promotes bolting

(3) A-Breaks seed and bud dormancy, B-Promotion of flowering in pineapple, C-Stress hormone, D-Rapid internode elongation in deep water rice plants

(4) A-initiates rooting in stem cutting, B-Promotes nutrient mobilisation, C-Promotes apical dominance, D-Promotion of seed maturation and dormancy

ZOOLOGY:

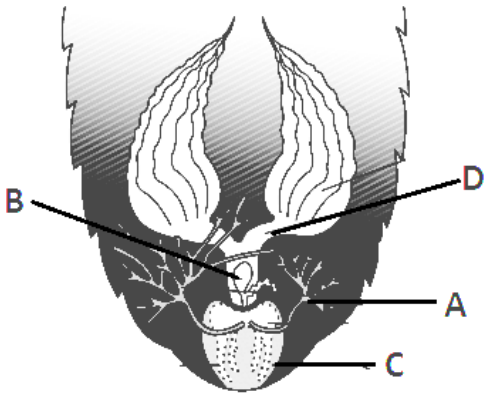
46. Which one is famous for hosting thousands of migratory birds coming from Siberia in winter

(1) Kanha national park
(2) Corbett national park
(3) Keoladeo national park
(4) Kaziranga national park

47. Comb jellies differ from jelly fishes in having

(1) radial symmetry and diploblastic condition
(2) comb plates and cnidocytes
(3) extra cellular digestion and intra cellular digestion
(4) comb plates and colloblasts

48. Given below is the figure of cockroach reproductive system. Select the option which gives the correct identification and its characteristic



(1) A-conglobate gland – secretes ootheca

(2) B-seminal vesicle – receives and stores the sperms

(3) C-vestibulum – posterior part of genital pouch

(4) D-vagina – into which open eight ovarioles

49. The second heart sound is associated with

(1) closure of semilunar valves during ventricular systole
(2) closure of semilunar valves during ventricular diastole
(3) closure of atrio-ventricular valves during ventricular diastole
(4) closure of atrio-ventricular valves during ventricular systole

50. Identify the wrong match.

(1) Bird flu – H_5N_1 virus

(2) Mule – Out breeding

(3) Edible fish – Pomfret

(4) Hisardale – Out crossing

51. During oogenesis, reduction division of meiosis is completed in

(1) tertiary follicle, forming ootid

(2) secondary follicle, forming secondary oocyte
(3) tertiary follicle, forming secondary oocyte
(4) Graafian follicle, forming ootid

52. Sixth amino acid in the beta globin chain of normal haemoglobin molecule is coded by codon

(1) G A G (2) G U G
(3) G A U (4) G U C

53. Bhimbetka rock shelter known for cave paintings by pre historic humans is located in

(1) Maharashtra (2) Meghalaya
(3) Uttar Pradesh (4) Madhya Pradesh

54. Natural aging of a lake by nutrient enrichment of its water is

(1) Biomagnification (2) Standing state
(3) Stratification (4) Eutrophication

55. Which of the following is incorrect about Pheretima

(1) Clitellar segments possess forests of integumentary nephridia

(2) Anterior loops and testes are present in same segments

(3) Segments without nephridia are also devoid of ganglia

(4) Tufts of pharyngeal nephridia and lymph glands are present in same segments

56. Total volume of air a person can inspire after a normal expiration is

(1) greater than VC

(2) lesser than EC

(3) greater than FRC

(4) equal to EC

57. Match the following.

Column-I	Column-II
A) Deltoid ridge	1) Ulna
B) Odontoid process	2) Scapula
C) Olecranon process	3) Humerus
D) Acromion process	4) Sternum
	5) Axis

	A	B	C	D
(1)	3	5	1	4
(2)	1	5	3	2
(3)	3	5	1	2
(4)	3	1	5	2

58. The symptoms of allergy can be quickly reduced by using

(1) Histamine, Epinephrine

(2) Heparin, Serotonin

(3) Dopamine, Histamine

(4) Adrenaline, Steroids

59. In a tubectomised person

(1) ovulation does not occur

(2) ovum can not be fertilized

(3) menstruation does not occur

(4) fimbriae can not collect ova

60. Triploblastic monoecious animal is

(1) Hirudinaria (2) Ancylostoma

(3) Laccifer (4) Scypha

61. Montreal protocol was signed in the year

(1) 1981 (2) 1985

(3) 1987 (4) 1989

62. Adhering junctions

(1) facilitate cell communication by connecting cytoplasm

(2) perform cementing to keep neighbouring cells together

(3) help to stop substances from leaking across a tissue

(4) facilitate rapid transfer of ions and molecules between the cells

63. pCO_2 in the alveoli of lung is

(1) equal to that in systemic arteries

(2) more than that in systemic veins

(3) more than that in tissues

(4) equal to that in pulmonary artery

64. Identify the correct match

(1) Myasthenia gravis – failure of release of neuro transmitter

(2) Heart failure – heart stops beating

(3) Muscular dystrophy – degeneration and paralysis of smooth muscles

(4) Atherosclerosis – narrowing of lumen of coronary arteries

65. Which of the following is a depressant and slows down the body functions

(1) Crack (2) Pep pills

(3) Smack (4) Charas

66. Identify the correct set of peptide hormones

(1) Epinephrine, aldosterone, glucagon

(2) Insulin, adrenaline, cortisol

(3) Secretin, thyrocalcitonin, estradiol

(4) Erythropoietin, cholecystokinin, thymosin

67. Which of the following prevents conception by blocking the entry of sperms through the cervix

(1) Implants (2) Lippes loop

(3) Vaults (4) Progestasert

68. Females have same number of chromosomes as males and produce two types of ova in

(1) Grasshoppers (2) Birds

(3) Fumea (4) Both 2 and 3

69. Connell's field experiments were in support of

(1) competitive exclusion

(2) interference competition

(3) competitive release

(4) co-existence

70. Which of the following is a correct match of the organism and its two characters.

(1) Lancelet – cartilaginous vertebral column, ventral heart

(2) Limulus – metamerism, book gills

(3) Lamprey – scales are absent, fins are absent

(4) Loligo – tube feet, calcareous ossicles

71. In frog, testis is attached to the kidney by

(1) mesovarium (2) mesometrium

(3) mesorchium (4) mesosalpinx

72. Choose the wrong match related to kidney

(1) selective secretion of K^+ and H^+ ions – collecting duct

(2) active or passive transport of electrolytes – ascending limb of Henle's loop

(3) selective secretion of HCO_3^- and Na^+ ions – proximal convoluted tubule

(4) conditional reabsorption of Na^+ and water – distal convoluted tubule

73. Melatonin is derived from

(1) Threonine (2) Tyrosine

(3) Histidine (4) Tryptophan

74. A woman homozygous for incontinentia pigment is married to a normal man. In the progeny

(1) only sons are affected

(2) only daughters are affected

(3) half of the progeny are normal

(4) all the children are affected

75. In disruptive natural selection, more individuals acquire

(1) peripheral character value at one end of the distribution curve

(2) mean character value

(3) value other than the mean character value

(4) peripheral character value at both ends of distribution curve

76. Formation of tropical forests needs mean annual temperature and mean annual precipitation as

(1) 10 – 23^0C and 100 – 200 cm

(2) 5 – 15^0C and 100 – 250 cm

(3) 18 – 25^0C and 150 – 400 cm

(4) 15 – 25^0C and 50 – 150 cm

77. Which of the following is an amphibian

(1) Ichthyosaur (2) Hydrophis

(3) Ichthyornis (4) Ichthyophis

78. Which of the following options best represents the enzyme composition of pancreatic juice

(1) Trypsinogen, chymotrypsinogen, lipase, amylase

(2) Trypsinogen, enterokinase, chymotrypsin, lipase

(3) Trypsin, carboxy peptidase, nuclease, amylase

(4) Trypsinogen, amylase, nucleosidase, lipase

79. Which of the following is not a part of renal pyramid of the kidney

(1) loop of Henle

(2) convoluted tubules

(3) vasa recta

(4) collecting duct

80. Nerve impulses from thoraco-lumbar division of autonomic nervous system causes

(1) contraction of circular muscles of iris

(2) increased secretion of digestive juices

(3) constriction of bronchi

(4) contraction of arrector pili muscles

81. Choose the incorrect match.

(1) Von Baer – embryos never pass through the adult stages of other animals

(2) Oparin & Haldane – formation of life was preceded by chemical evolution

(3) Special creation theory – age of earth is 4.5 billion years

(4) Hugo devries – mutations are random and directionless

82. One of the key concepts of Darwinian theory of evolution is

(1) inheritance of acquired characters

(2) genetic drift

(3) heritable mutation

(4) branching descent

83. Extinction of cichlid fish species in lake Victoria was due to

(1) over - exploitation

(2) co-extinction

(3) habitat loss

(4) Alien species invasion

84. Identify the incorrect match.

(1) Torpedo – electric organ

(2) Pristis – Placoid scales

(3) Trygon – Poison sting

(4) Pterophyllum – Claspers

85. In intestine, chylomicron formation takes place in

(1) lumen of the small intestine

(2) lacteals of the villi

(3) epithelial cells of intestinal mucosa

(4) blood capillaries of villi

86. In the internal ear, hair cells that act as auditory receptors are located in

(1) Scala tympani (2) Scala vestibuli

(3) Sacculus (4) Scala media

87. How many among the following are bacterial diseases

a) Diphtheria b) Leprosy

c) Polio d) Pertussis

e) Rabies f) Tetanus

g) Plague

(1) four (2) five

(3) six (4) seven

88. Which of the following is incorrect about Sertoli cells of testis

(1) act as phagocytes consuming the residual cytoplasm during spermatogenesis

(2) secretes Anti Mullerian hormone during early stages of foetal life

(3) have LH receptors on their membranes

(4) secretes androgen binding protein

89. Hepatic portal vein transports the blood to

(1) stomach (2) intestine

(3) liver (4) spleen

90. The relation between species richness and area $S = CA^z$ is a

(1) straight line

(2) rectangular hyperbola

(3) hypobola

(4) sigmoid

PHYSICS

91. The ends of a stretched wire of length L are fixed at $x = 0$ and $x = L$. In one experiment, the displacement of the wire is $y_1 = A\sin(\pi x/L)\sin\omega t$ and energy is E_1, and in another experiment its displacement is $y_2 = A\sin(2\pi x/L)\sin 2\omega t$ and energy is E_2. Then

(1) $E_2 = E_1$

(2) $E_2 = 2E_1$

(3) $E_2 = 4E_1$

(4) $E_2 = 16E_1$

92. An open tube is in resonance with string (frequency of vibration of tube is n_0). If tube is dipped in water so that 75% of length of tube is inside water, then the ratio of the frequency of tube to string now will be

(1) 1

(2) 2

(3) 2/3

(4) 1/2

93. 'Mirage' is a phenomenon due to

(1) Diffraction of light

(2) Refraction of light

(3) Total internal reflection of light

(4) Doppler effect

94. A luminous object is placed at a distance of 30 cm from the convex lens of focal length 20 cm. On the other side of the lens, at what distance from the lens a convex mirror of radius of curvature 10 cm be placed in order to have an upright image of the object coincident with it

(1) 12 cm

(2) 30 cm

(3) 50 cm

(4) 60 cm

95. Two coherent monochromatic light beams of intensities I and 4I are superposed. The maximum and minimum possible intensities in the resulting beam are

(1) 5I and I

(2) 5I and 3I

(3) 9I and I

(4) 9I and 3I

96. When the momentum of a proton is changed by an amount P_0, the corresponding change in the de-Broglie wavelength is found to be 0.25%. Then, the original momentum of the proton was

(1) $400\,P_0$

(2) $100\ P_0$

(3) P_0

(4) $4\ P_0$

97. Work function of lithium and copper are respectively 2.3 eV and 4.0 eV. Which one of the metal will be useful for the photoelectric cell working with visible light ? ($h = 6.6 \times 10^{-34}$ J-s, $c = 3 \times 10^8$ m/s)

(1) Lithium

(2) Copper

(3) both

(4) none of the above

98. Which of the following statements about the Bohr model of the hydrogen atom is false

(1) Acceleration of electron in n = 2 orbit is less than that in n = 1 orbit

(2) Angular momentum of electron in n = 2 orbit is more than that in n = 1 orbit

(3) Kinetic energy of electron in n = 2 orbit is less than that in n = 1 orbit

(4) Potential energy of electron in n = 2 orbit is less than that in n = 1 orbit

99. γ -rays radiation can be used to create electron-positron pair. In this process of pair production, γ-rays energy cannot be less than

(1) 5.0 MeV

(2) 4.02 MeV

(3) 15.0 MeV

(4) 1.02 MeV

100. N atoms of a radioactive element emit n alpha particles per second. The half life of the element is(in sec)

(1) $\frac{n}{N}$

(2) $\frac{N}{n}$

(3) $\frac{0.693\,N}{n}$

(4) $\frac{0.693\,n}{N}$

101. An AM wave has 1800 watt of total power content, For 100% modulation, the carrier should have power content equal to

(1) 1000 watt

(2) 1200 watt

(3) 1500 watt

(4) 1600 watt

102. If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be

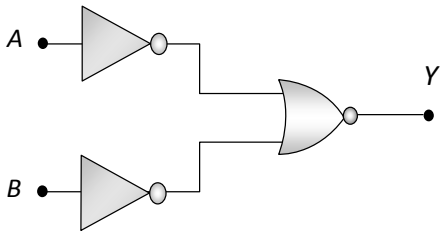
(1) 50 Hz

(2) 70.7 Hz

(3) 100 Hz

(4) 25 Hz

103. Which logic gate is represented by the following combination of logic gates



(1) OR

(2) NAND

(3) AND

(4) NOR

104. A spherical conductor of radius 2m is charged to a potential of 120 V. It is now placed inside another hollow spherical conductor of radius 6m. Calculate the potential to which the bigger sphere would be raised.

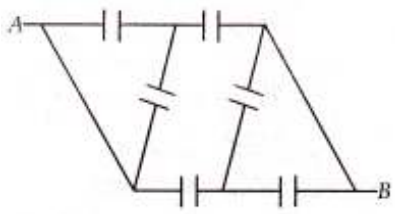
(1) 20 V

(2) 40 V

(3) 60 V

(4) 80 V

105. A network of six identical capacitors, each of value C is made as shown in the figure. Equivalent capacitance between points A and B is



(1) C/4

(2) 3C/4

(3) 4C/3

(4) 3C

106. A source of e.m.f. E = 15 V and having negligible internal resistance is connected to a variable resistance so that the current in the circuit increases with time as i = (1.2 t + 3). Then, the total charge that will flow in first five second will be

(1) 10 C

(2) 20 C

(3) 30 C

(4) 40 C

107. The resistance of 10 metre long potentiometer wire is 1ohm/meter. A cell of e.m.f. 2.2 volts and a high resistance box are connected in series to this wire. The value of resistance taken from resistance box for getting potential gradient of 2.2 millivolt/metre will be

(1) 790Ω

(2) 810Ω

(3) 990Ω

(4) 1000Ω

108. A voltmeter of resistance 1000Ω gives full scale deflection when a current of 100 mA flow through it. The shunt resistance required across it to enable it to be used as an ammeter reading 1 A at full scale deflection is

(1) 10000Ω

(2) 9000Ω

(3) 222Ω

(4) 111Ω

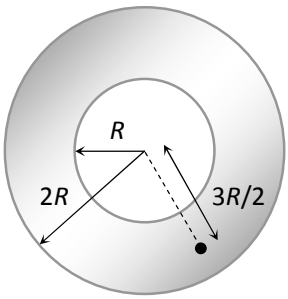
109. Figure shows the cross-sectional view of the hollow cylindrical conductor with inner radius 'R' and outer radius '2R', cylinder carrying uniformly distributed current i along its axis. The magnetic induction at point 'P' at a distance 3R/2 from the axis of the cylinder will be

(1) Zero

(2) $\frac{5\mu_0 i}{72\pi R}$

(3) $\frac{7\mu_0 i}{18\pi R}$

(4) $\frac{5\mu_0 i}{36\pi R}$



110. A player with 3 m long iron rod runs towards east with a speed of 30 km/hr. Horizontal component of earth's magnetic field is 4×10^{-5} Wb/m^2. If he is running with rod in horizontal and vertical positions, then the potential difference induced between the two ends of the rod in two cases will be

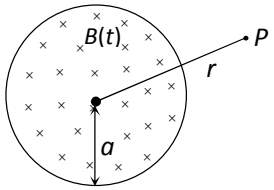
(1) Zero in vertical position and $1\times10^{-3}\,V$ in horizontal position

(2) $1\times10^{-3}\,V$ in vertical position and zero is horizontal position

(3) Zero in both cases

(4) $1\times10^{-3}\,V$ in both cases

111. A uniform but time-varying magnetic field B(t) exists in a circular region of radius 'a' and is directed into the plane of the paper, as shown. The magnitude of the induced electric field at point P at a distance r from the centre of the circular region



(1) Is zero

(2) Decreases as $\frac{1}{r}$

(3) Increases as r

(4) Decreases as $\frac{1}{r^2}$

112. In the hysteresis cycle, the value of H needed to make the intensity of magnetisation zero is called

(1) Retentivity (2) Coercive force

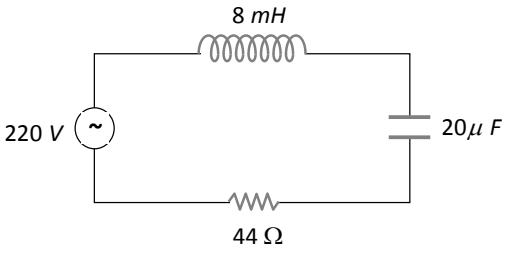
(3) Lorentz force (4) eddy effect

113. If a magnetic needle in a dip circle is suspended at an angle 30° to the magnetic meridian, it makes an angle of 45° with the horizontal. The real dip is

(1) $\tan^{-1}(\sqrt{3}/2)$ (2) $\tan^{-1}(\sqrt{3})$

(3) $\tan^{-1}(\sqrt{3/2})$ (4) $\tan^{-1}(2/\sqrt{3})$

114. For the series LCR circuit shown in the figure, what is the resonance frequency and the amplitude of the current at the resonating frequency



(1) $2500\,rad-s^{-1}$ and $5\sqrt{2}\,A$

(2) $2500\,rad-s^{-1}$ and 5A

(3) $2500\,rad-s^{-1}$ and $\frac{5}{\sqrt{2}}A$

(4) $25\,rad-s^{-1}$ and $5\sqrt{2}\,A$

115. **The position x of a particle with respect to time t along x-axis is given by $x = 9t^2 - t^3$ where x is in metres and t in seconds. The position of this particle when it achieves maximum speed along the +x direction will be at**

(1) 54 m

(2) 81 m

(3) 24 m

(4) 32 m

116. **A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90°. The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is**

(1) $\frac{ML^2}{6}$

(2) $\frac{\sqrt{2}ML^2}{24}$

(3) $\frac{ML^2}{24}$

(4) $\frac{ML^2}{12}$

117. **A particle is rotating in anti clock wise direction in xy-plane about the Z-axis. Its angular momentum about the centre of rotation is directed along**

1) –ve Z-axis 2) +ve Z-axis

3) –ve Y- axis 4) +ve Y-axis

118. **A particle P is moving in a circle of radius 'a ' with a uniform speed v. C is the centre of the circle and AB is a diameter. When passing through B, the angular velocity of P about A and C are in the ratio**

(1) 1 : 1

(2) 2 : 1

(3) 1 : 2

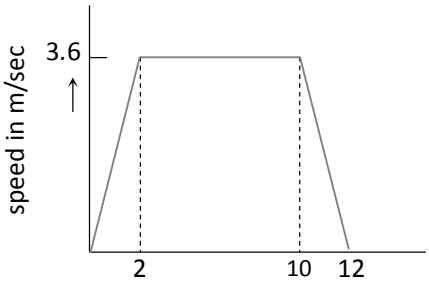
(4) 4 : 1

119. **A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity u at an angle $\left(\frac{\pi}{2} - \theta\right)$ with the horizontal, it reaches maximum height H_2. The relation between the horizontal range R of the projectile, H_1 and H_2 is**

(1) $R = 4\sqrt{H_1 H_2}$ (2) $R = 4(H_1 - H_2)$

(3) $R = 4(H_1 + H_2)$ (4) $R = \frac{H_1^2}{H_2^2}$

120. **A lift is going up. The total mass of the lift and the passenger is 1500 kg. The variation in the speed of the lift is as given in the graph. The tension in the rope pulling the lift at t = 11th sec will be**



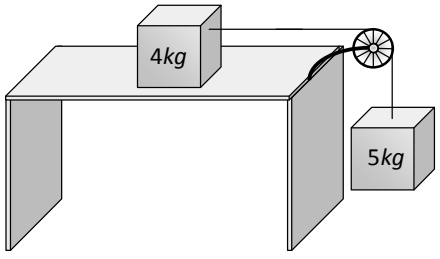
(1) 17400 N

(2) 14700 N

(3) 12000 N

(4) Zero

121. **Two masses of 4 kg and 5 kg are connected by a string passing through a frictionless pulley and are kept on a frictionless table as shown in the figure. The acceleration of 5 kg mass is**



(1) $49\,m/s^2$ (2) $5.44\,m/s^2$

(3) $19.5\,m/s^2$ (4) $2.72\,m/s^2$

122. A body of mass 4 kg moving with velocity 12 m/s collides with another body of mass 6 kg at rest. If two bodies stick together after collision, then the loss of kinetic energy of system is

(1) Zero

(2) 288 J

(3) 172.8 J

(4) 144 J

123. A pump motor is used to deliver water at a certain rate from a given pipe. To obtain twice as much water from the same pipe in the same time, power of the motor has to be increased to

(1) 16 times

(2) 4 times

(3) 8 times

(4) 2 times

124. Which of the following physical quantities do not have the same dimensions?

1)Pressure and stress

2)Tension and surface tension

3)Strain and angle

4)Energy and work

125. If v_e and v_o represent the escape velocity and orbital velocity of a satellite corresponding to a circular orbit of radius R, then

(1) $v_e = v_o$

(2) $\sqrt{2}\, v_o = v_e$

(3) $v_e = v_0 / \sqrt{2}$

(4) $v_e = v_0 / 2$

126. A large horizontal surface moves up and down in SHM with an amplitude of 1 cm. If a mass of 10 kg (which is placed on the surface) is to remain continually in contact with it, the maximum frequency of S.H.M. will be

(1) 0.5 Hz

(2) 1.5 Hz

(3) 5 Hz

(4) 10 Hz

127. The breaking stress of a wire depends upon

(1) Material of the wire

(2) Radius of the wire

(3) length of the wire

(4) all the above

128. A large number of water drops each of radius r combine to have a drop of radius R. If the surface tension is T and the mechanical equivalent of heat is J, then the rise in temperature will be(For water, take specific heat,s = 1 and density, d = 1)

(1) $\frac{2T}{rJ}$

(2) $\frac{3T}{RJ}$

(3) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

(4) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

129. The velocity of kerosene oil in a horizontal pipe is 5 m/s. If $g = 10 m/s^2$ then the velocity head of oil will be

(1) 1.25 m

(2) 12.5 m

(3) 0.125 m

(4) 125 m

130. A uniform metal rod is used as a bar pendulum. If the room temperature rises by 10°C, and the coefficient of linear expansion of the metal of the rod is 2×10^{-6} per °C, the period of the pendulum will have percentage increase of

(1) -2×10^{-3}

(2) -1×10^{-3}

(3) 2×10^{-3}

(4) 1×10^{-3}

131. If there are no heat losses, the heat released by the condensation of x gm of steam at 100°C into water at 100°C can be used to convert y gm of ice at 0°C into water at 100°C. Then the ratio y : x is nearly

(1) 1 : 1

(2) 5 : 2

(3) 1 : 3

(4) 3 : 1

132. According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surroundings, and n is equal to

(1) 4

(2) 3

(3) 2

(4) 1

133. The molar heat capacity in a process of a diatomic gas if it does a work of Q/4 when a heat of Q is supplied to it is

(1) $\frac{2}{5}R$

(2) $\frac{5}{2}R$

(3) $\frac{10}{3}R$

(4) $\frac{6}{7}R$

134. A Carnot's engine used first an ideal monoatomic gas then an ideal diatomic gas. If the source and sink temperature are 411^oC and 69^oC respectively and the engine extracts 1000 J of heat in each cycle, then area enclosed by the PV diagram is

(1) 100 J

(2) 300 J

(3) 500 J

(4) 700 J

135. The molecules of a given mass of a gas have r.m.s velocity of $200ms^{-1}$ at 27^oC and $1.0x10^5Nm^{-2}$ pressure. When the temperature and pressure of the gas are respectively. 127^oC and $0.05x10^5Nm^{-2}$, the r.m.s. velocity of its molecules in ms^{-1} is:

1) $100\sqrt{2}$ 2) $\frac{400}{\sqrt{3}}$

3) $\frac{100\sqrt{2}}{3}$ 4) $\frac{100}{3}$

CHEMITSRY

136. The edge lengths of the unit cells in terms of the radius of spheres constituting fcc, bcc and simple cubic unit cell are, respectively.

(1) $2\sqrt{2}r, \frac{4r}{\sqrt{3}}, 2r$ (2) $\frac{4r}{\sqrt{3}}, 2\sqrt{2}r, 2r$

(3) $2r, 2\sqrt{2}r, \frac{4r}{\sqrt{3}}$ (4) $2r, \frac{4r}{\sqrt{3}}, 2\sqrt{2}r$

137. In the cubic close packing, the unit cell has _____

(1) Four tetrahedral voids, each of which is shared by four adjacent unit cells.

(2) Four tetrahedral voids, without the unit cell.

(3) Eight tetrahedral voids, each of which is shared by four adjacent unit cells.

(4) Eight tetrahedral voids within the unit cells

138. Value of Henry's law constant K_H

(1) Increases with increase in temperature

(2) Decreases with increase in temperature

(3) Remains constant

(4) First increases then decreases

139. In comparison to a 0.01 M solution of glucose, the depression in freezing point of a 0.01M $MgCl_2$ solution is

(1) The same

(2) About twice

(3) About three times

(4) About six times

140. $\wedge_M^o(H_2O)$ **is equal to**

(1) $\wedge_M^o(HCl)-\wedge_M^o(NaOH)+\wedge_M^o(NaCl)$

(2) $\wedge_M^o(HNO_3)+\wedge_M^o(NaNO_3)-\wedge_M^o(NaOH)$

(3) $\wedge_M^o(HNO_3)+\wedge_M^o(NaOH)+\wedge_M^o(NaNO_3)$

(4) $\wedge_M^o(NH_4OH)+\wedge_M^o(HCl)-\wedge_M^o(NH_4Cl)$

141. In the electrolysis of aqueous NaCl solution, which of the half-cell reaction will occur at anode ?

(1) $Na^+ + e^- \rightarrow Na$, $E^o = -2.71V$

(2) $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$, $E^o = +1.23V$

(3) $H^+ + e^- \rightarrow \frac{1}{2}H_2$, $E^o = 0V$

(4) $Cl^- \rightarrow \frac{1}{2}Cl_2 + e^-$, $E^o = +1.36V$

142. Which of the following expressions is correct for the rate of reaction given below ?

$5Br^- + BrO_3^- + 6H^+ \rightarrow 3Br_2 + 3H_2O$

(1) $\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[H^+]}{\Delta t}$

(2) $\frac{\Delta[Br^-]}{\Delta t} = \frac{6}{5}\frac{\Delta[H^+]}{\Delta t}$

(3) $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$

(4) $\frac{\Delta[Br^-]}{\Delta t} = \frac{6\Delta[H^+]}{\Delta t}$

143. Which of the following will show Tyndall effect ?

(1) Aqueous solution of soap below CMC

(2) Aqueous solution of soap above CMC

(3) Aqueous solution of NaCl

(4) Aqueous solution of sugar

144. Which of the following electrolytes will have maximum coagulating value for Ag I/Ag^+ sol ?

(1) Na_2S (2) Na_3PO_4

(3) Na_2SO_4 (4) NaCl

145. Auto reduction can be used for the extraction of

(1) Iron (2) Aluminium

(3) Zinc (4) Copper

146. Which of the following is isoelectronic pair ?

(1) ICl_2, ClO (2) $BrO_2^-, BrF2^+$

(3) BrF, ClO_2 (4) CN^-, O_3

147. Which of the following oxides of nitrogen is a mixed anhydride of two acids ?

(1) N_2O (2) NO

(3) N_2O_4 (4) N_2O_3

148. On heating with sulphur, Na_2SO_3 produces

(1) $Na_2S_2O_3$ (2) Na_2S_5

(3) $Na_2S_2O_7$ (4) $Na_2S_4O_6$

149. Which of the following is a high-spin complex ?

(1) $[Co(NH_3)6]^{3+}$ (2) $[Fe(CN)6]^{4-}$

(3) $[CoF6]^{3-}$ (4) $[Zn(NH_3)_6]^{2+}$

150. The IUPAC name of $[Co(NH_3)_5(ONO)]SO_4$ is

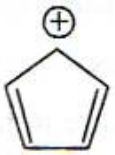
(1) Pentaammine nitrito – O – cobalt (III) sulphate

(2) Pentamine nitrito – O – cobalt (III) sulphate

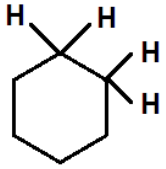
(3) Pentaammine nitrito – N – cobalt (III) sulphate

(4) Pentaammine nitro – O – cobaltate (III) sulphate

151. Which of the following is expected to be aromatic ?

(1) 

(2) 

(3) 

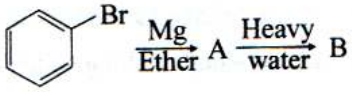
(4) 

152. The cyclic polymerization of methyl acetylene gives mesitylene. The number of sigma bonds in mesitylene is

(1) 16 (2) 12

(3) 18 (4) 21

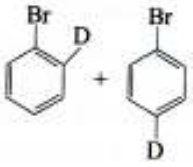
153. The final product in the following reaction is



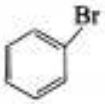
(1) 

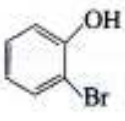
(2) 

(3) 

(4) 

154. The product obtained when phenol reacts with HBr is

(1) 

(2) 

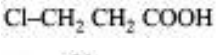
(3) 

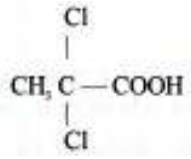
(4) No reactions

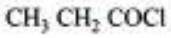
155. The product obtained in the reaction

$CH_3CH_2COOH + Cl_2 \xrightarrow{\text{Red P}}$ **is**

(1) 

(2) 

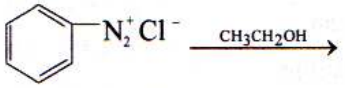
(3) 

(4) 

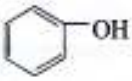
156. The reaction of acetamide with NaOBr in an alkaline medium produces

(1) CH_3NH_2 (2) NH_3

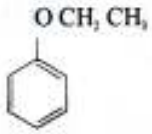
(3) $CH_3CH_2NH_2$ (4) CH_3CN



157. compound 'A' is

(1) 

(2) 

(3) 

(4) 

158. Which of the following B group vitamins can be stored in our body ?

(1) B_1 (2) B_2

(3) B_6 (4) B_{12}

159. Which of the following reactions of glucose can be explained only by its cyclic structure ?

(1) Glucose forms penta acetate

(2) Glucose reacts with hydroxylamine to form an oxime

(3) Penta acetate of glucose does not react with hydroxylamine

(4) Glucose is oxidized by HNO_3 to gluconic acid

160. The commercial name of polyacrylonitrile is
(1) Dacron (2) Orlon
(3) Bakelite (4) Nylon-6

161. A narrow – spectrum antibiotic is active against
(1) Gram positive or gram negative bacteria
(2) Gram negative bacteria only
(3) Single organism or one disease
(4) Both gram positive and gram negative bacteria

162. Which of the following d-block element has lowest enthalpy of sublimation ?
(1) Iron (2) Zinc
(3) Chromium (4) Copper

163. The ions of metals of group 12 (Zn, Cd, Hg) have a complete d-subshell and so
(1) Behave like super conductor
(2) Behave like semiconductor
(3) are not transition metals
(4) Are very high melting solids

164. The configuration $1s^2\ 2s^2\ 2p^5\ 3s^1$ is valid for the
(1) Ground state of fluorine
(2) Excited state of fluorine
(3) Excited state of neon
(4) Excited state of sodium

165. To which of the following is Bohr's theory applicable ?
(1) He^+ (2) Li^{2+}
(3) H (4) All

166. Among the following, the molecule that is linear is
(1) CO_2 (2) NO_2
(3) SO_2 (4) ClO_2

167. Which of the following molecule has the least dipole moment ?
(1) NF_3 (2) CH_4
(3) SO_2 (4) NH_3

168. Which of the following gaseous mixture is dalton's law of partial pressure not applicable ?
(1) Ne + He + SO_2
(2) NH_3 + HC*l*
(3) $O_2 + N_2 + CO_2$
(4) $N_2 + H_2 + O_2$

169. Which of the following contains the greatest number of atoms ?
(1) 1.0 g Butane (C_4H_{10})
(2) 1.0 g Nitrogen N_2
(3) 1.0 g Silver (Ag)
(4) 1.0 g of Water (H_2O)
(At mass: H = 1 N = 14, O = 16, C = 12, Ag = 108)

170. Dimer of ICl_3 is an orange solid, what is the hybridization of I in it ?
(1) sp^3 (2) sp^3d
(3) sp^3d^2 (4) sp^3d^3

171. The value of $\Delta_f H^o$ for NH_3 is -91.8 KJ/mol. Calculate the enthalpy change for the reaction

$$NH_3(g) \rightarrow \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

(1) -91.8 KJ/mol (2) -18.6 KJ/mol
(3) +91.8 KJ/mol (4) +183.6 KJ/mol

172. The order of screening effect of electrons of s, p, d and f orbital of a given shell of an atom on its outer shell electron is
(1) s > p > d > f (2) f > d > p > s
(3) p > d > s > f (4) f > p > s > d

173. At constant pressure, upon the addition of helium at the equilibrium $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ the degree of dissociation of
(1) PCl_5 will decrease
(2) PCl_5 will increase
(3) PCl_3 will increase
(4) Cl_2 will increase

174. The solubility of AgCl in 0.1 M NaCl will
(1) Increase as compared to pure solution
(2) Decrease as compared to pure solution
(3) Remain unchanged
(4) AgC*l* will dissociate completely

175. The solubility of a salt A_2B_2 is 1.0×10^{-3} M. Its solubility product is

(1) 1.08×10^{-13} (2) 1.08×10^{-15}

(3) 1.08×10^{-10} (4) 1.08×10^{-17}

176. The pH of 10^{-10} M H_2SO_4 will be almost

(1) 4 (2) 7

(3) 6 (4) 0

177. Which of the following is electron – precise hydride ?

(1) B_2H_6 (2) NH_3

(3) H_2O (4) SiH_4

178. Which of the following metal carbonate is thermally least stable ?

(1) $MgCO_3$ (2) $CaCO_3$

(3) $SrCO_3$ (4) $BaCO_3$

179. Organosilicon polymers are called silicones. The chain length of silicone polymer can be controlled by adding

(1) CH_3SiCl_3 (2) $(CH_3)_2SiCl_2$

(3) $(CH_3)_3SiCl$ (4) $(CH_3)_3Si$

180. Photochemical smog occurs is warm, dry and suny climate. One of the following is not amongst the components of photochemical smog. Which component is this ?

(1) NO_2

(2) O_3

(3) SO_2

(4) Unsaturated hydrocarbon

SRI CHAITANYA EDUCATIONAL INSTITUTIONS, INDIA

NEET GRAND TEST – 10 KEY

SR.ELITE , AIIMS-60,SR MPL&MEDICON **Date: 07-02-2019**

BOTANY

1) **1**	2) **4**	3) **4**	4) **2**	5) **4**	6) **2**	7) **3**	8) **2**	9) **3**	10) **4**
11) **3**	12) **2**	13) **3**	14) **4**	15) **4**	16) **2**	17) **2**	18) **4**	19) **2**	20) **2**
21) **4**	22) **3**	23) **1**	24) **4**	25) **2**	26) **1**	27) **3**	28) **2**	29) **3**	30) **2**
31) **1**	32) **2**	33) **2**	34) **2**	35) **3**	36) **2**	37) **1**	38) **4**	39) **2**	40) **2**
41) **3**	42) **4**	43) **3**	44) **2**	45) **2**					

ZOOLOGY

46) **3**	47) **4**	48) **3**	49) **2**	50) **4**	51) **3**	52) **1**	53) **4**	54) **4**	55) **4**
56) **3**	57) **3**	58) **4**	59) **2**	60) **1**	61) **3**	62) **2**	63) **1**	64) **4**	65) **3**
66) **4**	67) **3**	68) **2**	69) **3**	70) **2**	71) **3**	72) **3**	73) **4**	74) **4**	75) **4**
76) **3**	77) **4**	78) **1**	79) **2**	80) **4**	81) **3**	82) **4**	83) **4**	84) **4**	85) **3**
86) **4**	87) **2**	88) **3**	89) **3**	90) **2**					

PHYSICS

91) **3**	92) **2**	93) **3**	94) **3**	95) **3**	96) **1**	97) **1**	98) **4**	99) **4**	100) **3**
101) **2**	102) **3**	103) **3**	104) **2**	105) **3**	106) **3**	107) **3**	108) **4**	109) **4**	110) **2**
111) **2**	112) **2**	113) **1**	114) **2**	115) **1**	116) **4**	117) **2**	118) **3**	119) **1**	120) **3**
121) **2**	122) **3**	123) **3**	124) **2**	125) **2**	126) **3**	127) **1**	128) **3**	129) **1**	130) **4**
131) **4**	132) **4**	133) **3**	134) **3**	135) **2**					

CHEMISTRY

136) **1**	137) **4**	138) **1**	139) **3**	140) **4**	141) **2**	142) **3**	143) **2**	144) **4**	145) **4**
146) **2**	147) **3**	148) **1**	149) **3**	150) **1**	151) **4**	152) **3**	153) **3**	154) **4**	155) **1**
156) **1**	157) **2**	158) **4**	159) **3**	160) **2**	161) **1**	162) **2**	163) **3**	164) **3**	165) **4**
166) **1**	167) **2**	168) **2**	169) **1**	170) **3**	171) **3**	172) **1**	173) **2**	174) **2**	175) **1**
176) **2**	177) **4**	178) **1**	179) **3**	180) **3**					

Physics Solutions

91. Energy (E) $\propto$ (Amplitude)^2 (Frequency)^2

Amplitude is same in both the cases, but frequency 2ω in the second case is two times the frequency (ω) in the first case. Hence $E_2 = 4E_1$

92. For open tube, $n_0 = \frac{v}{2l}$

For closed tube length available for resonance is

$l' = l \times \frac{25}{100} = \frac{l}{4}$

$\therefore$ Fundamental frequency of water filled tube $n = \frac{v}{4l'} = \frac{v}{4 \times (l/4)} = \frac{v}{l} = 2n_0 \Rightarrow \frac{n}{n_0} = 2$

93. theory point

94. For lens $u = 30\ cm,\ f = 20\ cm$, hence by using $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{+20} = \frac{1}{v} - \frac{1}{-30} \Rightarrow v = 60\ cm$

The final image will coincide the object, if light ray falls normally on convex mirror as shown. From figure it is seen clear that separation between lens and mirror is 60 – 10 = 50 *cm.*

95. $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{I} + \sqrt{4I})^2 = 9I\ \ I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2 = (\sqrt{I} - \sqrt{4I})^2 = I$

96. $\lambda \propto \frac{1}{p} \Rightarrow \frac{\Delta p}{p} = -\frac{\Delta\lambda}{\lambda} \Rightarrow \left|\frac{\Delta p}{p}\right| = \left|\frac{\Delta\lambda}{\lambda}\right|$

$\Rightarrow \frac{p_0}{p} = \frac{0.25}{100} = \frac{1}{400} \Rightarrow p = 400\, p_0$.

97. From $\lambda_0 = \frac{12375}{W_0}$

The maximum wavelength of light required for the photoelectron emission, $(\lambda_0)_{Li} = \frac{12375}{2.3} = 5380\ \text{Å}$.

Similarly $(\lambda_0)_{Cu} = \frac{12375}{4} = 3094$ Å.

Since the wavelength 3094 Å does not in the visible region, but it is in the ultraviolet region. Hence to work with visible light, lithium metal will be used for photoelectric cell.

98. As n increases P.E. also increases.

99. Energy released by $\gamma-$ rays for pair production must be greater than 1.02 *MeV*

100. $\frac{dN}{dt} = -\lambda N \Rightarrow n = -\lambda N$ (Given $\frac{dN}{dt} = n$)

$\therefore \lambda = -\frac{n}{N}$ $\therefore$ Half life $= \frac{0.693}{\lambda} = \frac{0.693}{\lambda} = \frac{0.693\ N}{n}$ sec

101. $P_t = P_c\left(1 + \frac{m_a^2}{2}\right)$; Here $m_a = 1$

$\Rightarrow 1800 = P_c\left(1 + \frac{(1)^2}{2}\right) \Rightarrow P_c = 1200\, W$

102. Theory point

103. $Y = \overline{\overline{A} + \overline{B}}$

According to De morgan's theorem

$Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = A.B$

This is the output equation of 'AND' gate.

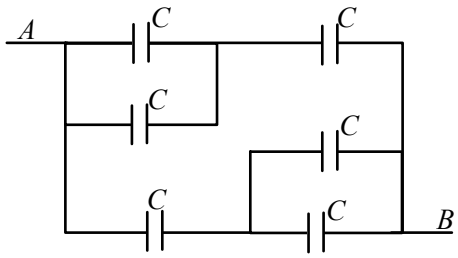
104. If charge acquired by the smaller sphere is Q then it's potential $120 = \frac{kQ}{2}$ ..... (i)

Also potential of the outer sphere

$V = \frac{kQ}{6}$ .....(ii)

From equation (i) and (ii) $V = 40$ *volt*

105. Eq.circuit



$$\Rightarrow C_{eq} = \frac{2C}{3} + \frac{2C}{3} = \frac{4C}{3}$$

106. $i = \frac{dQ}{dt} \Rightarrow dQ = idt \Rightarrow Q = \int_{t_1}^{t_2} i\,dt = \int_0^5 (1.2t + 3)dt$

$$= \left[\frac{1.2t^2}{2} + 3t\right]_0^5 = 30\,C$$

107. Potential gradient $x = \frac{V}{L} = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L}$

$$\Rightarrow 2.2 \times 10^{-3} = \frac{2.2}{(10 + R_h)} \times 1 \Rightarrow R' = 990\ \Omega$$

108. By using $\frac{i}{i_g} = 1 + \frac{G}{S}$

$$\Rightarrow \frac{i}{100 \times 10^{-3}} = 1 + \frac{1000}{S} \Rightarrow S = \frac{1000}{9} = 111\ \Omega$$

109. By using $B = \frac{\mu_0 i}{2\pi r}\left(\frac{r^2 - a^2}{b^2 - a^2}\right)$ here $r = \frac{3R}{2},\ a = R,\ b = 2R$

$$\Rightarrow \quad B = \frac{\mu_0 i}{2\pi\left(\frac{3R}{2}\right)} \times \left\{\frac{\left(\frac{3R}{2}\right)^2 - R^2}{(2R^2) - R^2}\right\} = \frac{5.\mu_o i}{36\pi R}.$$

110. If player is running with rod in vertical position towards east, then rod cuts the magnetic field of earth perpendicularly (magnetic field of earth is south to north).

Hence Maximum emf induced is

$$e = Bvl = 4 \times 10^{-5} \times \frac{30 \times 1000}{3600} \times 3 = 1 \times 10^{-3}\ volt$$

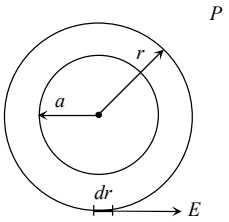
When he is running with rod in horizontal position, no field is cut by the rod, so $e = 0$

111. Construct a concentric circle of radius r. The induced electric field (E) at any point on the circle is equal to that at P. For this circle

$$\oint \vec{E}.d\vec{l} = \left|\frac{d\phi}{dt}\right| = A\left|\frac{dB}{dt}\right|$$

or $E \times (2\pi r) = \pi a^2 . \left|\frac{dB}{dt}\right|$

$$\Rightarrow E = \frac{a^2}{2r}\left|\frac{dB}{dt}\right| \Rightarrow E \propto \frac{1}{r}$$



112. theory point

113. Let the real dip be ϕ, then $\tan\phi = \frac{B_V}{B_H}$

For apparent dip,

$$\tan\phi' = \frac{B_V}{B_H \cos\beta} = \frac{B_V}{B_H \cos 30^o} = \frac{2B_V}{\sqrt{3}B_H}$$

or $\tan 45^o = \frac{2}{\sqrt{3}} . \tan\phi$ or $\phi = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

114. Resonance frequency

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 10^{-3} \times 20 \times 10^{-6}}} = 2500\ rad/\sec$$

Resonance current = $\frac{V}{R}=\frac{220}{44}=5A$

115. $V=\frac{dx}{dt}=\frac{d}{dt}\left(gt^2-t^3\right)=18t-3t^2$,

For maximum speed

$\frac{dv}{dt}=0\Rightarrow 18-6t=0$

$\therefore\ t=3\sec$

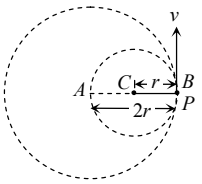
$\therefore\ X_{\max}=81-27=54m$

116. **I=** $2\times\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2\times\frac{1}{3}=\frac{ML^2}{12}$.

117. Theory point

118. Angular velocity of particle *P* about point *A,*

$\omega_A=\frac{v}{r_{AB}}=\frac{v}{2r}$



Angular velocity of particle *P* about point *C*,

$\omega_C=\frac{v}{r_{BC}}=\frac{v}{r}$

Ratio $\frac{\omega_A}{\omega_C}=\frac{v/2r}{v/r}=\frac{1}{2}$.

119. $H_1=\frac{u^2\sin^2\theta}{2g}$ and $H_2=\frac{u^2\sin^2(90-\theta)}{2g}=\frac{u^2\cos^2\theta}{2g}$

$H_1H_2=\frac{u^2\sin^2\theta}{2g}\times\frac{u^2\cos^2\theta}{2g}=\frac{(u^2\sin 2\theta)^2}{16g^2}=\frac{R^2}{16}$

$\therefore\ R=4\sqrt{H_1H_2}$

120. At 11th second lift is moving upward with acceleration

$a=\frac{0-3.6}{2}=-1.8m/s^2$

Tension in rope, $T=m(g-a)$

$=1500\,(9.8-1.8)=12000\ N$

121. $a=\frac{m_2}{m_1+m_2}\times g=\frac{5}{4+5}\times 9.8=\frac{49}{9}=5.44\ m/s^2$

122. Loss in K.E. = $\frac{m_1m_2}{2(m_1+m_2)}(u_1-u_2)^2$

$=\frac{4\times 6}{2\times 10}\times(12-0)^2=172.8\,J$

123. Power of a pump $=\frac{1}{2}\rho Av^3$

To get twice amount of water from same pipe *v* has to be made twice. So power is to be made 8 times

124. dimensional formulae

125. $v_e=\sqrt{2gR}$ and $v_0=\sqrt{gR}\ \therefore\ \sqrt{2}\,v_0=v_e$

126. For body to remain in contact $a_{\max}=g$

$\therefore\omega^2A=g\Rightarrow 4\pi^2n^2A=g$

$\Rightarrow n^2=\frac{g}{4\pi^2A}=\frac{10}{4(3.14)^2\,0.01}=25\ \Rightarrow n=5\ Hz$

127. theory point

128. Rise in temperature, $\Delta\theta=\frac{3T}{JSd}\left(\frac{1}{r}-\frac{1}{R}\right)$

$\therefore \Delta\theta = \frac{3T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$

129. Velocity head $h = \frac{v^2}{2g} = \frac{(5)^2}{2 \times 10} = 1.25\ m$

130. Fractional change in period

$$\frac{\Delta T}{T} = \frac{1}{2}\alpha\Delta\theta = \frac{1}{2} \times 2 \times 10^{-6} \times 10 = 10^{-5}$$

$$\% \text{ change} = \frac{\Delta T}{T} \times 100 = 10^{-5} \times 100 = 10^{-3}\ \%$$

131. Heat released to convert x *gm* of steam at 100°*C* to water at 100°*C* is $x \times 540$ *cals*.

If y *gm* of ice is converted from 0°*C* to water at 100°*C* it requires heat $y \times 80 + y \times 1 \times 100 = 180\ y$

$\therefore \quad x \times 540 = 180\ y$ or $\frac{y}{x} = \frac{540}{180} = \frac{3}{1}$

132. According to Newton's law

Rate of cooling $\propto$ temperature difference $\Delta\theta$

133. $dU = C_V dT = \left(\frac{5}{2}R\right)dT$ or $dT = \frac{2(dU)}{5R}$ .....(i)

From first law of thermodynamics

$dU = dQ - dW = Q - \frac{Q}{4} = \frac{3Q}{4}$. Now molar heat capacity $C = \frac{dQ}{dT} = \frac{Q}{\frac{2(dU)}{5R}} = \frac{5RQ}{2\left(\frac{3Q}{4}\right)} = \frac{10}{3}R$

134. $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{(273 + 69)}{(273 + 411)} = 0.5$

$\Rightarrow$ Work done $= \eta \times Q = 0.5 \times 1000 = 500\ J$

.

135. Rms velocity depends on temperature only. $\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$

CHEMISTRY

1. $a = 2r$ for simple cubic

   $\sqrt{2}\ a, = 4r$ for fcc

   $\sqrt{3}\ a, = 4r$ for bcc

2. In ccp, the octahedral voids are present at the center of each edge as well as at the center of cube.

   While tetrahedral voids are at the body diagonal and each body diagonal has two tetrahedral voids. So, total eight tetrahedral voids are present per unit cell.

3. $p_1 = K_H x_i$

   The solubility of gas decreases with increase in temperature. Therefore, K_H increases with increase in temperature.

4. $i = 3$ for $MgCl_2$, therefore, ΔT_f of 0.01 M $MgCl_2$ will be three times than 0.01 M glucose solution.

5. $\wedge_M^0(H_2O) = \lambda_{H+}^0 + \lambda_{OH^-}^0$

   $= \wedge_{M(HCl)}^0 + \wedge_{M(NH_4OH)}^0 - \wedge_{M(NH_4CL)}^0$

6. $E°H_2O/O_2 < E° cl^-/cl_2$. Therefore, H_2O gets oxidized at anode to liberate O_2 gas.

7. $\frac{-1}{5}\frac{\Delta[Br^-]}{\Delta t} = \frac{-1}{6}\frac{\Delta[H^+]}{\Delta t}$

   $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$.

8. Aqueous solution of soap above CMC is an associated colloid and hence shows Tyndall effect.

9. Ag $1/Aq^e$ is coagulated by negatively charged ion and higher the charge, higher the coagulating power and lower will be the coagulating value.

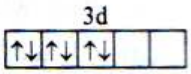
10. $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$

11. BrO_2^- and BrF_2^+ is an example of isoelectronic pair.

12. N_2O_4 on reaction with water gives a mixture of nitrous acid and nitric acid.

    $N_2O_4 + H_2O \rightarrow HNO_2 + HNO_3$

13. $Na_2SO_3 + S \rightarrow Na_2S_2O_3$
    (Sodium thioshulphate)

14. $[Co(NH_3)_6]^{3+}$ CO^{3+} (Ar) 

    NH_3 is a strong ligand.

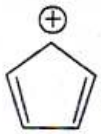
$[Fe(CN)_6]^{4-}$ Fe^{2+} (Ar) 3d ↑↓ ↑↓ ↑↓ | | CN^- is a strong ligand.

$[CoF_6]^{3-}$ CO^{3+} (Ar) 3d ↑↓ | ↑ | ↑ | ↑ | ↑ F^- is a weak ligand

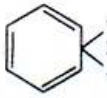
$[Zn(NH_3)_6]^{2+}$ Zn^{2+} (Ar) 3d ↑↓ ↑↓ ↑↓ ↑↓ ↑↓

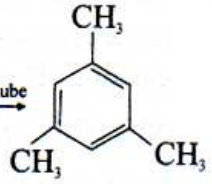
as this complex has the maximum number of unpaired electron.

15. The correct IUPAC name is given

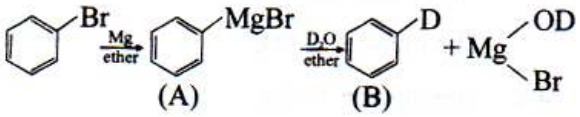
16.  is non-aromatic as it is not satisfying Huckel's rule.

 is also non-aromatic as it is not satisfying Huckel's rule

 In this, one of the carbon is sp^3, hence non-aromatic.

17. $CH_3 - C \equiv CH$ (Methyl acetylene) $\xrightarrow{\text{Red hot tube}}$  (Mesitylene)

It has 21 sigma bonds.

18. 

19.  + HBr → No reaction as C – O bond has a partial double bond character due to resonance.

20. This is an example of HVZ reaction. . Excess of Cl_2 can replace more than one alpha hydrogen.

$$CH_3CH_2COOH + Cl_2 \xrightarrow{\text{Red P}} CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - COOH$$

21. This is Hofmann bromamide reaction

$$\underset{\text{acetamide}}{CH_3CONH_2} \xrightarrow{NaOBr} CH_3NH_2 + NaBr + Na_2CO_3 + H_2O$$

22. CH_3CH_2OH acts as a reducing agent and reduces benzene diazenium chloride to benzene.

23.

24. The cyclic structure of glucose does not have a free aldehyde group and hence penta acetate of glucose does not form oxime.

25. $$n\,CH_2 = \underset{\displaystyle CN}{\underset{|}{CH}} \longrightarrow \underset{\text{Polyacrylonitrite}}{\left\{CH_2 - \underset{\displaystyle CN}{\underset{|}{CH}}\right\}}$$

It other name is orlon or acrilan.

26.

27. Zinc due to its fully filled $3d^{10}$ has weak metallic bonding and weak interatomic forces of attraction, hence lowest enthalpy of sublimation.

28. d-block metals having incompletely filled d-subshell or one of their ions have incompletely filled d-subshell are called transition metals.

29. $_{10}Ne\ 1s^2\ 2s^2\ 2p_6$

Its excited state configuration is $1s^2\ 2s^2\ 2p^5\ 3s^1$

30. Bohr's theory is applicable to atom/ion having only one electron.

31. NO_2, SO_2 and ClO_2 all three have bent structure. CO_2 is linear.

32.

CH_4 is non polar

33. Doltan's law is applicable to mixture of chemically non-reacting gases.

as NH_3 and HCl, HBr reacts to NH_4Cl and NH_4Br^3.

34. $1.0\ g\ C_4H_{10} \rightarrow \frac{1}{58}\ mol\ C_4H_{10} = 14 \times \frac{1}{58}\ mol$ atom = 0.24 mol atom

$1.0\ g\ N_2 \rightarrow \frac{1}{28}\ mol\ N_2 = 2 \times \frac{1}{28}\ mol$ atom

$= \frac{1}{14}\ mol\ atom = 0.07\ mol\ atom$

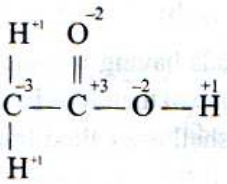
$1.0\ g\ Ag \rightarrow \frac{1}{108}\ mol\ Ag\ atom$

$= 0.009\ mol\ atom$

$1.0\ g\ H_2O \rightarrow \frac{1}{18}\ mol\ H_2O = \frac{3}{18}\ mol\ atom$

$= \frac{1}{6}\ mol\ atom = 0.167\ mol\ atom$

35.



36. $N_2\ (g) + 3\ H_2(g) \rightarrow 2\ NH_3\ (g)$

$\Delta H^\circ = -91.8$ KJ/mol.

For $2\ NH_3\ (g) \rightarrow N_2\ (g) + 3\ H_2\ (g)\ \Delta_f h^\circ = 91.8$ KJ/mol

37.

38. The addition of helium shifts the equilibrium towards right hand side and more PCl_5 will dissociate. The degree of dissociation of PCl_5 will increase.

39. Due to common Cl^- ion, the solubility of AgCl will decrease.

40. $A_2B_3\ (s) \rightleftharpoons 2A^{3+}_{(aq)} + 3\ B^{2-}_{(aq)}$

$K_{sp} = [A^{3+}]^2\ [B^{2-}]^3 = (2S)^2\ (3S)^3 = 108\ S^5$

Where S is the solubility.

$K_{rp} = 108 \times (1.0 \times 10^{-3})^5 =$

$108 \times 10^{-15} = 1.08 \times 10^{-13}$

41. It is a very dilute solution.

$(H^+) = (H^+)H_2SO_4 + (H^+)H_2O \approx (H^+)H_2O \approx 10^{-7}\ M$

Hence, the pH = 7.0

42. SiH_4 is an electron-precise hydride as silicon has a complete octet without any lone pair of electron.

NH_3, H_2O → Electron-rich hydrides

B_2H_6 → Electron-deficient hydrides

43. Dnone the group 2, the thermal stability increases due to stabilisation of large cation by large anion. $MgCO_3$ is thermally least stable out of the four given.

44.

45.